Trek Nepal Website Proposal

Student Id	: 2431868
Student Name	: Prakriti Khadka
Section	: L5CG24
Supervisor	: Priyanka Aryal
Reader	: Uttam Acharya
Submitted on	: 2025-09-06
Word Count	:2930

Table of Contents

1. Problem Statement 1
2. Academic Questions 2
3. Aims of the Project 3
4. Objectives of the Project 3
5. Project as a Solution 4
6. Background of the Study 5
6.1. Literature review 5
6.2. Himalayan Glacier Trekking 5
6.3. Nepal Hiking Team 6
7. Functional Decomposition Diagram (FDD) and Detailed Explanation of Functionalities 7
7.1. Trekking Package Exploration 7
7.2. Booking System 7
7.3. Expense Tracking 8
7.4. Interactive Maps 8
7.5. Real-Time Weather Updates 8
7.6. Guide Profiles, Reviews, and Ratings 8
7.7. Community Blogs and Experience Sharing 8
7.8. Eco-Friendly Trekking Resources 8
7.9. Device Accessibility 8
7.10. User Authentication & Role Management 8
8. Software Development lifecycle 9
9. Tech Stack 10
10. Gantt Chart 11
11. Testing Methodologies 12
12. Limitations and conclusion 13
References 14

Table of Figures

Figure 1:Functional Decomposition..... 7
Figure 2:Gantt Chart 11

1. Problem Statement

Trekking in Nepal is one of the most sought-after adventure activities, yet planning and managing treks remains a challenging process for many travellers. Currently, trekkers face several issues such as lack of centralized information about trekking routes, difficulty in finding verified guides, manual and unreliable expense tracking, and limited access to real-time weather or altitude details. Many booking processes are still handled offline, leading to inefficiencies, miscommunication, and even fraudulent practices. Additionally, trekkers often struggle to estimate costs, compare packages, and manage resources effectively. These challenges result in a poor user experience and hinder Nepal's tourism potential.

There is a need for a modern, digital platform that can integrate trekking information, booking management, expense tracking, and guide interaction into a single system. By addressing these challenges, Trek Nepal aims to provide a reliable, user-friendly, and innovative solution that benefits both trekkers and service providers while promoting Nepal as a global trekking destination.

2. Academic Questions

- How can a centralized web application improve the overall trekking experience in Nepal for both local and international trekkers?
- What digital features (e.g., booking, expense tracking, guide ratings, interactive maps) are most effective in addressing the current challenges faced by trekkers?
- How can expense tracking and budget estimation be integrated into a trekking platform to help travellers manage their financial resources more efficiently?
- To what extent can the use of interactive maps, real-time weather updates, and route information enhance the safety and preparedness of trekkers?
- How can the Trek Nepal platform support guides and service providers in reaching more clients and managing their services effectively?
- How does implementing responsive design features improve the accessibility and usability of the system across devices?

3. Aims of the Project

The main aim of the Trek Nepal web application is to provide an accessible, efficient, and engaging platform that enhances the trekking experience in Nepal by offering seamless planning, expense management, booking, and interaction with guides and service providers.

4. Objectives of the Project

The objectives of Trek Nepal are as follows:

- To create a user-friendly digital platform for exploring trekking packages across Nepal.
- To implement a booking and scheduling system for trekkers and guides.
- To integrate an expense tracker for better financial planning during treks.
- To provide interactive maps and real-time weather information for improved trek safety.
- To facilitate experience sharing through reviews, and ratings.
- To promote eco-friendly and sustainable trekking practices.
- To ensure cross-platform accessibility through responsive design and modern web technologies.

5. Project as a Solution

To address the challenges faced by trekkers, guides, and service providers in Nepal's trekking industry, the proposed solution is the development of a comprehensive web application named Trek Nepal. This system aims to integrate all essential trekking-related services into a single digital platform that is reliable, accessible, and user-friendly. The solution provides a centralized platform where users can explore detailed trekking packages, compare routes, and access real-time information about weather and altitude conditions. By integrating live weather forecasts and updates, the system enhances safety and preparedness, enabling trekkers to plan their journeys around changing mountain weather conditions. A booking and scheduling system implemented to replace manual processes, allowing trekkers to reserve packages, guides, and services efficiently. To assist with financial planning, the system incorporates an expense tracker and budget estimator that enables users to log and monitor expenses throughout their trek. From a technical standpoint, the system is designed with responsive design principles and Progressive Web App (PWA) features, ensuring accessibility across both desktop and mobile devices. These features allow trekkers to continue accessing vital data such as weather updates, maps, and saved bookings even in areas with limited connectivity. Through these features, the Trek Nepal application addresses the inefficiencies of the current trekking system by eliminating manual methods, improving safety with real-time weather insights, enhancing user experience, and supporting both trekkers and service providers in a single, modernized platform.

6. Background of the Study

6.1.Literature review

Paper 1: Trek Monitoring System: Enhancing Safety and Adventure in the Outdoors using Arduino UNO

Existing research highlights the importance of integrating health monitoring, location tracking, and weather forecasting to ensure safety in trekking and other outdoor activities. Wearable and portable devices using sensors such as temperature and pulse have been widely studied for real-time health assessment, while GPS-based tracking has been shown to improve navigation and enable timely rescue operations in remote areas. Studies further emphasize the role of user-friendly web platforms that provide route information, weather updates, and continuous monitoring, thereby assisting both trekkers and trekking companies. Overall, the literature supports that integrated monitoring systems significantly enhance trekking safety by addressing health risks, environmental uncertainties, and communication gaps.

Paper2: Development of an Indicator System for Evaluating Risks in Trekking Tourism

Trekking tourism has gained importance as a recreational activity that helps reduce stress and connect people with nature. However, due to its adventurous nature, risk assessment is a vital aspect of ensuring safety. Previous studies have applied structured methods like the Delphi method and Analytic Hierarchy Process (AHP) to evaluate risks in outdoor tourism. Grounded in theories such as the Systems Theory of Accident Causation and the Human Factors Theory, research highlights the need to identify key risk factors and assign proper weights to evaluation criteria. Findings across the literature suggest that scientifically designed risk assessment frameworks provide reliable tools for tourism stakeholders, organizations, and authorities to enhance safety management in trekking activities.

Paper3: Recommender Systems for Outdoor Adventure Tourism Sports: Hiking, Running and Climbing

Adventure tourism, encompassing activities such as hiking, running, and climbing, is a rapidly expanding sector that has attracted significant interest from both tourists and technology developers. Recent studies highlight the growing role of Artificial Intelligence in enhancing outdoor experiences, particularly through recommender systems that provide personalized suggestions based on user preferences. While substantial progress has been made in developing such systems for running and hiking, other adventure sports remain underexplored. Systematic reviews of decision support systems demonstrate the need for standardized methodologies to guide the design of AI-based solutions in this field. Overall, the literature emphasizes recommender systems as a key technology for advancing personalization, safety, and user engagement in adventure tourism.

6.2.Himalayan Glacier Trekking

Himalayan Glacier Trekking is one of the leading private trekking and tour operators in Nepal. Their website offers detailed trekking packages, itineraries, and cost breakdowns, along with options to make online inquiries or bookings. While the platform provides comprehensive trek information, it is primarily business-oriented and focuses on package sales. It lacks user-cantered features such as real-time weather updates, integrated expense tracking, or customizable trekking plans. The absence of a centralized platform that combines both planning and financial management leaves trekkers dependent on external tools.

6.3.Nepal Hiking Team

The Nepal Hiking Team website is another established trekking operator portal offering trekking packages across major destinations like Everest, Annapurna, and Langtang. It provides itineraries, price details, and booking functionality. However, like other operator websites, it is limited to commercial purposes. It does not incorporate digital features such as expense management, trek safety alerts, weather forecasting, or user-generated content. (Team, 2025)

7. Functional Decomposition Diagram (FDD) and Detailed Explanation of Functionalities

Link : https://www.figma.com/board/73LC8Wm69us3M0RKULBS4u/Untitled?node-id=0-1&p=f&t=GoDf1c8zq5Y7pxum-0

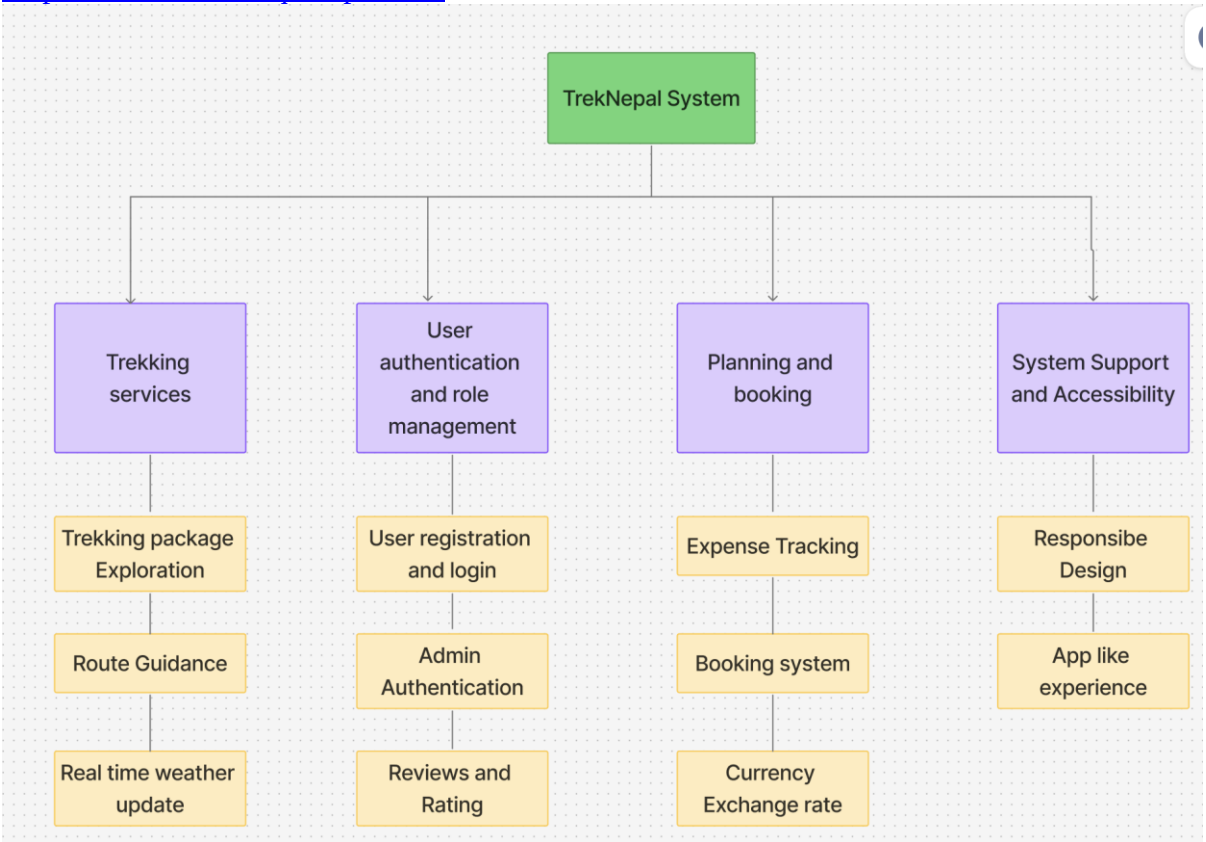

Figure 1:Functional Decomposition

A Functional Decomposition Diagram helps to break down the complex Trek Nepal system into smaller, manageable sub-tasks, enabling better understanding of its structure and functionality. The system is divided into core modules that collectively contribute to delivering a complete trekking management solution. Each feature is designed to address a specific challenge faced by trekkers, guides, or service providers.

7.1. Trekking Package Exploration

This feature allows users to browse, search, and filter available trekking packages across Nepal. It includes details such as routes, duration, difficulty level, altitude, cost, and itinerary. By centralizing trekking information, it simplifies trip planning and helps users make informed decisions.

7.2. Booking System

The booking system enables trekkers to reserve packages, guides, and services directly through the platform. It provides availability calendars, real-time confirmations, and secure management of reservations, reducing reliance on manual or offline processes.

7.3. Expense Tracking

This feature allows trekkers monitor expenses during their journey. It helps users visualize their expenses in the form of chart.

7.4. Interactive Maps

The system integrates map services to provide interactive trekking routes. This feature enhances trek preparation and navigation while ensuring safety through better awareness of terrain and routes.

7.5. Real-Time Weather Updates

By integrating weather APIs, the platform provides trekkers with live forecasts, alerts, and climate insights for specific trekking routes. This enhances safety and preparedness by helping trekkers adjust schedules based on mountain weather conditions.

7.6. Guide Profiles, Reviews, and Ratings

The platform hosts verified guide profiles that include experience details, availability, and reviews from past trekkers. This fosters trust, allows trekkers to choose reliable guides, and gives guides an opportunity to reach more clients.

7.7. Community Blogs and Experience Sharing

This feature provides trekkers with the ability to share their experiences, photos and reviews.

7.8. Eco-Friendly Trekking Resources

This feature educates users about waste management, responsible trekking, and preserving Nepal's natural and cultural heritage.

7.9. Device Accessibility

The platform is designed as a Progressive enabling cross-platform usability on both desktop and mobile devices. Features such as offline access to saved routes, cached weather data, and booking details ensure continuous usability even with limited internet connectivity.

7.10. User Authentication & Role Management

The system includes secure user authentication with account registration, login, and password recovery. Different roles are supported, including trekkers, guides, and administrators. Administrators manage platform settings, while guides and service providers handle their packages and client bookings.

8. Software Development lifecycle

The development of the Trek Nepal web application follows a structured Software Development Lifecycle (SDLC) to ensure systematic planning, design, implementation, testing, and deployment. (Atlassian, 2024) SDLC provides a framework for delivering a high-quality software product that meets user requirements and project objectives. The following phases were adopted for Trek Nepal:

i. **Requirement Analysis:** In this phase, the functional and non-functional requirements of the system were gathered through research, observation of existing trekking platforms in Nepal, and interviews with potential users. Key requirements included trekking package management, booking systems, expense tracking, real-time weather updates, interactive maps, and guide interactions.

ii. **System Design:** Based on the requirements, system architecture, database schema, and user interface design were created. This phase involved designing the functional decomposition diagram (FDD), wireframes, and overall system workflow. The design ensures modularity, scalability, and a user-friendly interface.

iii. **Implementation (Coding):** The system was developed using HTML, CSS, JavaScript for the frontend and Python with Django for the backend. Django's built-in database (SQLite) was used to store data securely. Features such as booking, expense tracking, interactive maps, and weather API integration were implemented in this phase.

iv. **Testing:** After implementation, the system underwent rigorous testing using unit testing, integration testing, functional testing, usability testing, performance testing, and security testing. Any identified bugs were fixed, and system performance was optimized to ensure reliability and smooth operation.

v. **Deployment:** Once testing was complete and the system was stable, Trek Nepal was deployed to a live environment, making it accessible to users on both desktop and mobile devices.

vi. **Maintenance:** System will undergo regular maintenance to fix bugs, update features, integrate new APIs, and ensure long-term sustainability. Feedback from users will guide future enhancements to improve usability and functionality.

9. Tech Stack

The Trek Nepal web application is developed using a combination of modern web technologies and project management tools to ensure a reliable, scalable, and user-friendly system. The chosen tech stack supports core features such as trekking package management, booking, expense tracking, interactive maps, and real-time weather updates.

i. **Frontend:** The frontend of Trek Nepal is built using HTML, CSS, and JavaScript, which provide the structure, styling, and interactivity required for a responsive and engaging user interface.

ii. **Backend:** The backend is developed using Django, a high-level Python web framework that facilitates rapid development and clean, maintainable code. Django handles server-side logic, user authentication, data processing, and integration with frontend components efficiently.

iii. **Database:** The application uses Django's built-in database (SQLite) for storing user data, trekking packages, bookings, guides, and expense records. SQLite is lightweight and fully integrated with Django, making it suitable for rapid development and testing.

iv. **Project Management Tools:** Jira and Trello are utilized for project planning, task assignment, and tracking development progress. These tools enhance team collaboration and ensure efficient project execution.

v. **Version Control:** Git is used for version control, enabling code management, tracking changes, and collaboration among developers. (Chacon, n.d.)

10. Gantt Chart

A Gantt Chart is a project management tool that visually represents tasks and timelines using horizontal bars. It helps in planning, scheduling, and tracking project progress by showing task duration, dependencies, and deadlines. For Trek Nepal, the Gantt Chart ensures clear task allocation, better coordination, and timely project delivery. (Hirakata, 2022)

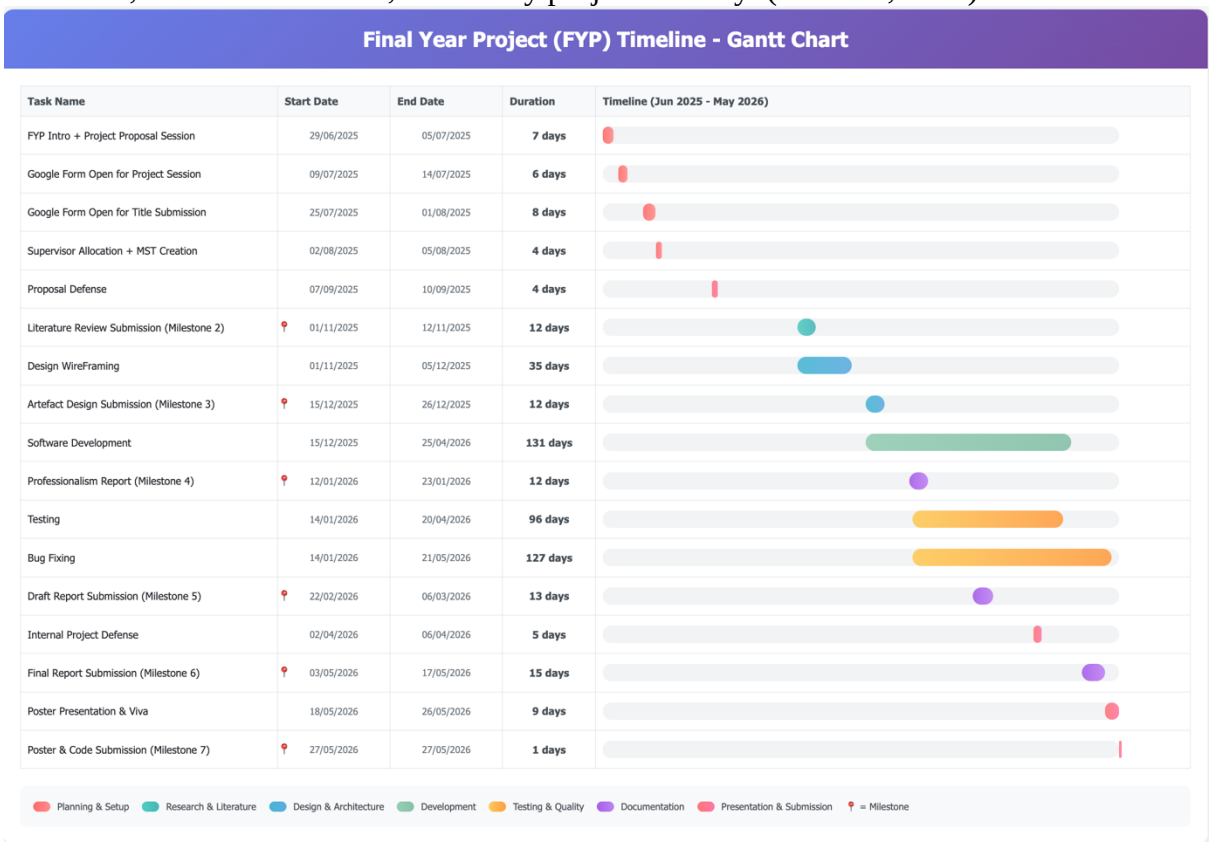

Figure 2:Gantt Chart

11. Testing Methodologies

Testing is a critical phase in the development of the Trek Nepal web application to ensure that all features function as intended, are reliable, and provide a smooth user experience. The testing process focuses on verifying system functionality, usability, performance, and security. (Anon., n.d.)

i. **Unit Testing:** Each individual component of the system, such as the booking module, expense tracker, and user authentication, is tested separately to ensure that it performs correctly. Unit testing helps identify and fix errors early in the development process.
ii. **Integration Testing:** After unit testing, different modules are combined, and integration testing is conducted to ensure that all system components work together seamlessly. For example, the integration of the booking system with the database and payment module is verified.
iii. Functional Testing: Functional testing evaluates whether the application meets its specified requirements. This includes testing core features such as trekking package browsing, guide interaction, expense tracking, interactive maps, and real-time weather updates.
iv. **Usability Testing:** The application is tested with real users to assess its user-friendliness, navigation, and interface design. Feedback from usability testing helps improve accessibility, responsiveness, and overall user experience.
v. **Performance Testing:** Performance testing measures the system's responsiveness and stability under various conditions, such as multiple simultaneous users or heavy data loads. This ensures that the application remains reliable and efficient even during peak usage.
vi. **Security Testing:** Security testing checks for vulnerabilities in the system, including data protection, secure login, and safe handling of sensitive user information. This is crucial to prevent unauthorized access and ensure user trust.
vii. **Regression Testing:** After updates or bug fixes, regression testing is performed to ensure that new changes do not negatively affect existing functionalities.

By employing these testing methodologies, Trek Nepal ensures a robust, secure, and user-friendly platform, capable of delivering a reliable trekking experience for users while minimizing potential errors or issues in real-world use.

12. Limitations and conclusion

Although the Trek Nepal web application introduces significant improvements in planning and managing treks in Nepal, it has certain limitations. The system depends on internet connectivity, which may be unreliable in remote trekking regions, limiting access to real-time weather updates and online bookings. Data accuracy for weather forecasts, altitude, and routes depends on external sources, and the current design is primarily suited for individual trekkers or small groups, posing scalability challenges. Payment gateway integration and offline functionality are also limited, and adoption by traditional trekking operators may be slow. Despite these constraints, Trek Nepal provides a comprehensive solution by integrating trekking packages, booking management, expense tracking, interactive maps, and live weather updates into a single platform, enhancing safety, efficiency, and the overall trekking experience. The system also promotes sustainable tourism and benefits both trekkers and service providers, demonstrating how digital transformation can modernize Nepal's trekking sector and support economic growth, while leaving room for future enhancements to address existing limitations.

References

Anon., n.d. *BrowserStack.* [Online]
Available at: https://www.browserstack.com/guide/software-testing-methodologies
[Accessed 14 Sep 2025].
Atlassian, 2024. *Atlassian.* [Online]
Available at: https://www.atlassian.com/agile/software-development/sdlc
[Accessed 14 Sep 2025].
Chacon, S. a. B. S., n.d. *GitHub.* [Online]
Available at: https://git-scm.com
[Accessed 14 Sep 2025].
Hirakata, V. O. M. G. V. e. a., 2022. Exploring the Gantt chart as a tool to highlight double report in case series published during the first wave of the COVID-19 pandemic. *https://doi.org/10.1186/s13643-022-02024-0*, pp. 100-500.
Team, N. H., 2025. *Nepal Hiking Team.* [Online]
Available at: https://www.nepalhikingteam.com/
[Accessed 14 Sep 2025].